

Does Liver Radiodensity Predict Chemotherapy Toxicity?

Time-to-event analysis across CAP, OXA, and CAPOX

Noah Bos & Abdullah Alam

2026-04-13

Study Objective

- Assess whether liver radiodensity (HU) predicts time to dose-limiting toxicity
- Outcome: number of cycles until toxicity occurs
- Adjusted for patient characteristics and hospital differences

Key takeaway: Liver HU is evaluated for its added predictive value in clinical decision-making.

Toxicity Endpoints Analyzed Separately

- CAP toxicity
- OXA toxicity
- CAPOX toxicity

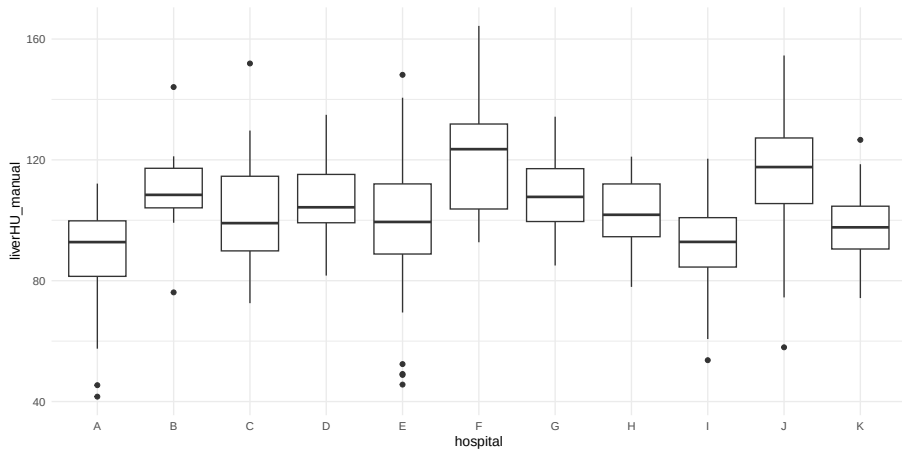
Key takeaway: Each treatment is evaluated using a separate time-to-event model.

Observed Toxicity Rates Differ by Treatment

Endpoint	N	Events	Censored
CAP	344	181	163
OXA	311	254	57
CAPOX	359	257	102

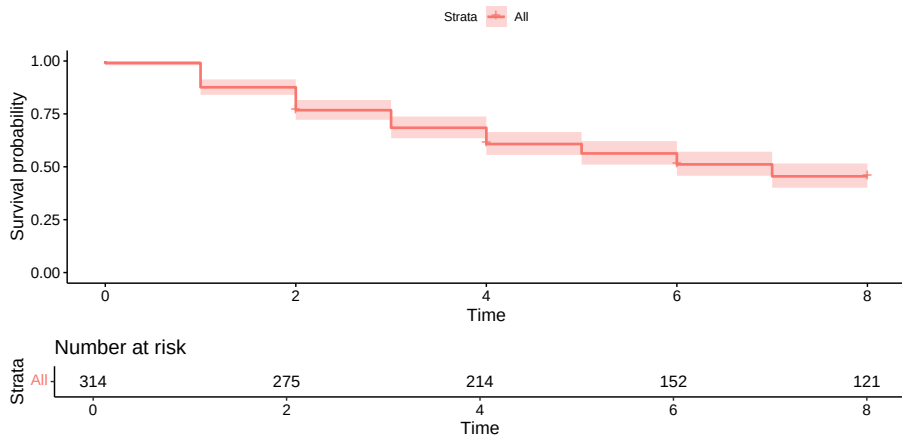
Key takeaway: Event rates differ across treatments, with sufficient events to support survival analysis.

Liver Radiodensity Varies Across Hospitals



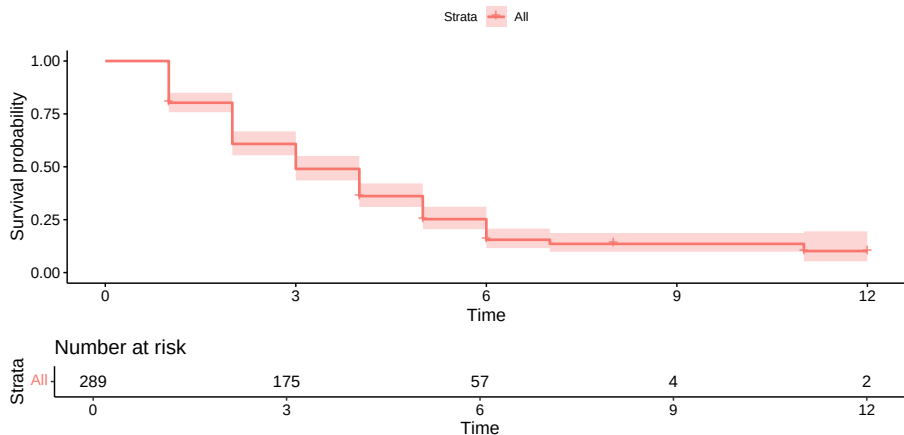
Key takeaway: Liver HU distributions differ between hospitals, suggesting baseline differences that are accounted for through stratification.

Time to Toxicity: CAP



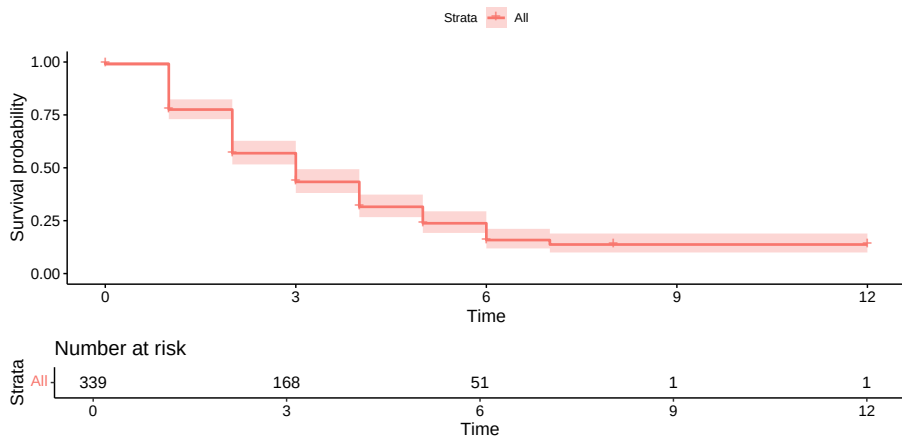
Key takeaway: The probability of remaining toxicity-free declines over cycles, with most events occurring early in treatment.

Time to Toxicity: OXA



Key takeaway: Toxicity risk is highest in early cycles. Estimates in later cycles are less reliable due to few patients remaining at risk.

Time to Toxicity: CAPOX



Key takeaway: A similar pattern is observed, with most toxicities occurring early and increasing uncertainty in later cycles.

Why Use Cox Models?

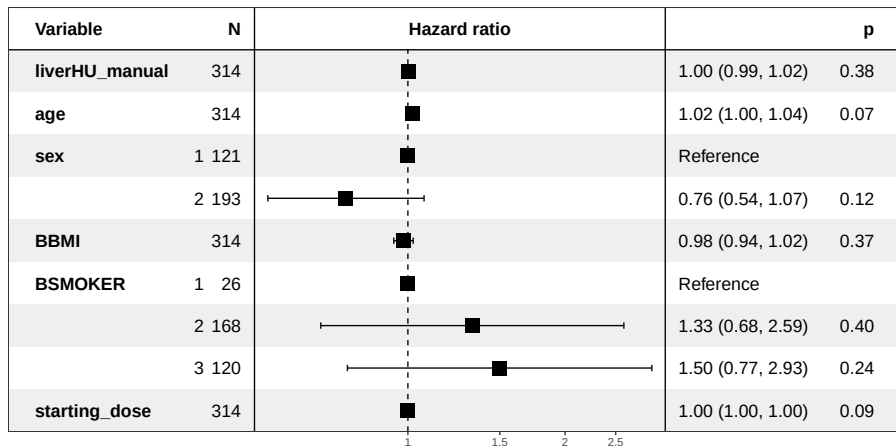
- Kaplan-Meier curves describe when toxicity occurs
- They do not quantify effects of predictors
- Cox models estimate the association between liver HU and toxicity
- Stratified by hospital to allow different baseline risks

Key takeaway: Cox models quantify whether liver radiodensity predicts toxicity after adjustment.

Association Between Liver HU and Toxicity

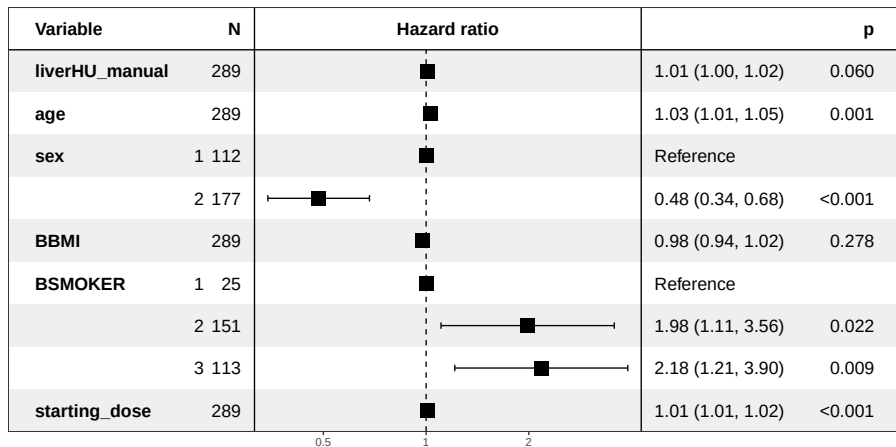
term	estimate	std.error	statistic	p.value	conf.low	conf.high	Model
liverHU_manual	1.00	0.01	0.88	0.38	0.99	1.02	CAP
age	1.02	0.01	1.79	0.07	1.00	1.04	CAP
sex2	0.76	0.18	-1.55	0.12	0.54	1.07	CAP
BBMI	0.98	0.02	-0.89	0.37	0.94	1.02	CAP
BSMOKER2	1.33	0.34	0.83	0.40	0.68	2.59	CAP
BSMOKER3	1.50	0.34	1.18	0.24	0.77	2.93	CAP
starting_dose	1.00	0.00	1.70	0.09	1.00	1.00	CAP
liverHU_manual	1.01	0.00	1.88	0.06	1.00	1.02	OXA
age	1.03	0.01	3.18	0.00	1.01	1.05	OXA
sex2	0.48	0.17	-4.15	0.00	0.34	0.68	OXA
BBMI	0.98	0.02	-1.09	0.28	0.94	1.02	OXA
BSMOKER2	1.98	0.30	2.30	0.02	1.11	3.56	OXA
BSMOKER3	2.18	0.30	2.61	0.01	1.21	3.90	OXA
starting_dose	1.01	0.00	3.59	0.00	1.01	1.02	OXA
liverHU_manual	1.01	0.00	1.22	0.22	1.00	1.02	CAPOX
age	1.01	0.01	0.90	0.37	0.99	1.02	CAPOX
sex2	0.75	0.14	-1.99	0.05	0.57	1.00	CAPOX
BBMI	0.99	0.02	-0.59	0.55	0.96	1.02	CAPOX
BSMOKER2	1.78	0.27	2.14	0.03	1.05	3.01	CAPOX
BSMOKER3	1.83	0.27	2.24	0.03	1.08	3.12	CAPOX

CAP: No Meaningful Association with Liver HU



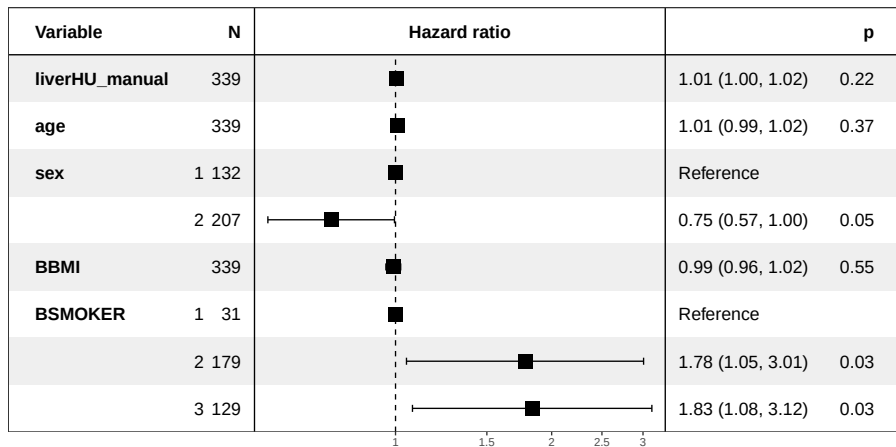
Key takeaway: The estimated effect of liver HU is close to null with no statistically significant association.

OXA: Small Trend, Not Statistically significant



Key takeaway: A slight association is observed, but the effect size is small and not significant

CAPOX: No Consistent Association Observed



Key takeaway: Estimates are close to null, with no clear evidence of an association.

Model Assumptions Hold

- No evidence of violation of the proportional hazards assumption
- Hazard ratios appear stable over time

	chisq	df	p
liverHU_manual	0.2255962	1	0.6348086
age	0.2120542	1	0.6451621
sex	0.7255615	1	0.3943257
BBMI	0.7826862	1	0.3763209
BSMOKER	0.1230178	2	0.9403446
GLOBAL	2.0444171	6	0.9155685

Key takeaway: The Cox model assumptions are satisfied, supporting the validity of the results.

Overall Interpretation

- Liver radiodensity shows no consistent or clinically meaningful association with time to toxicity
- Effect sizes are small and not statistically significant across endpoints
- Other clinical variables (e.g., age, smoking, dose) show stronger associations

Key takeaway: Liver radiodensity does not appear to add meaningful predictive value beyond standard clinical variables.

No clear association was observed when liver radio density was modeled continuously across its full range. A possible follow-up analysis would be to assess whether toxicity risk differs only at clinically meaningful thresholds or extreme radio density ranges.